Biomechanics of the Hand

Scott F.M. Duncan, MD, MPH, MBA[a,*], Caitlin E. Saracevic[b], Ryosuke Kakinoki, MD, PhD[c]

KEYWORDS

• Hand • Biomechanics • Thenar • Hypothenar • Interosseous • Lumbrical

KEY POINTS

- Ideally digits need sensation and freedom of motion to enable patients to use them effectively. The patients' perception of sensations and level of pain determines the eventual functional use and agility of the hand.
- Understanding how patients used their hand in daily living and employment will assist the surgeon in determining which procedure will be most effective to allow patients to return to their active pre-injury lifestyle and employment.
- Effective surgery ideally restores the biomechanical motions so that patients have optimum use.
- Optimum use can be restored through (1) existing anatomy or (2) tendon transfers, fusions, and even amputation.

INTRODUCTION

In its most simplified form, the hand is made up of a stable wrist with 2 digits, at the minimum, that are able to oppose against each other with some power. Ideally one or both of the digits are capable of motion so grasping can be performed. In its minimal form, one digit can be stable with one digit having motion to move against that stable post. Digits benefit from having sensation and from being pain free so that their usage is facilitated. In regard to biomechanical motion, the hand has 7 maneuvers that make up most hand functions.

1. The precision pinch, otherwise known as the terminal pinch, involves flexion of the interphalangeal (IP) joint of the thumb and the distal IP (DIP) joint of the index finger. The fingernail tips are brought together so that a small item, such as a pen, can be picked up (Fig. 1).
2. The next basic function is the oppositional pinch, otherwise known as the subterminal pinch. This pinch is where the pulp of the thumb and index finger are brought together with the IP and DIP joints in extension, which allows for increased forces to be generated through thumb opposition. It also relies on the first dorsal interosseous contracting while, simultaneously, the index profundus flexion is occurring (Fig. 2).
3. Key pinch maneuvering, in this situation, is when the thumb is adducted to the radial aspect of the index finger's middle phalanx. The key pinch maneuver does require a stable post, which in this situation is really the index finger. It also requires adequate length of the digit and a metacarpal phalangeal joint (MCP), which is capable of resisting thumb adduction (Fig. 3).
4. The chuck grip, otherwise known as the directional grip, allows the index finger, long finger, and thumb to come together to envelop a cylindrical object. A rotational and axial force is usually applied to the object when using this type of grip (Fig. 4).

[a] Department of Orthopedic Surgery, Ochsner Health System, 1514 Jefferson Highway, New Orleans, LA 70121, USA; [b] York University, 4700 Keele Street, Toronto, Ontario M3J 1P3, Canada; [c] Department of Orthopedic Surgery, Graduate School of Medicine, Kyoto University, 54 Shogoin Kawahara-cho, Sakyo-ku, Kyoto 606-8570, Japan
* Corresponding author.
E-mail address: Duncan.academic@gmail.com

Hand Clin 29 (2013) 483–492
http://dx.doi.org/10.1016/j.hcl.2013.08.003
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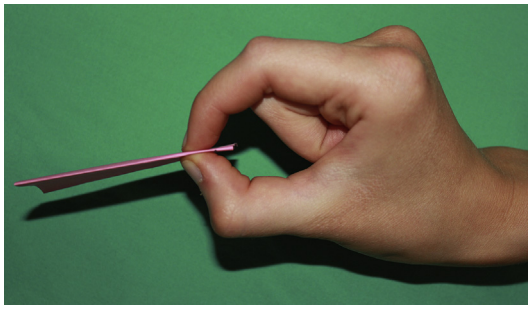

Fig. 1. Terminal pinch/precision pinch.

5. The hook grip requires finger flexion at the IP joints and extension at the MCP joints. This grip is used, for example, when one picks up a suitcase or a briefcase. It does not require thumb function (Fig. 5).
6. In the power grasp position, the fingers are flexed and the thumb is flexed and opposed relative to the other digits such as gripping a club or bat (Fig. 6).
7. The span grasp maneuver is when the DIP joints and the proximal IP (PIP) joints flex to approximately 30° and the thumb is palmarly abducted such that forces are generated between the thumb and fingers. This maneuver differs from the power grasps maneuver whereby forces are generated between the fingers and the palm. Stability is needed at the thumb, MCP, and IP joints. This type of grip is used, for example, to grab a ball (Fig. 7).

After surgery, the hand's ability to accommodate these various positions and exert the appropriate biomechanical forces through them determines how well patients recover from the injuries and procedures. For this reason, it is important to obtain a thorough history of patients to be able to emphasize which functions are mostly needed in order for patients to resume activities of daily living as well as previous employment.

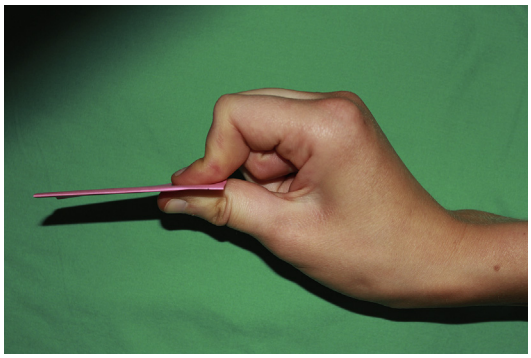

Fig. 2. Opposition pinch.

Fig. 3. Key pinch.

THE THUMB

The thumb is a unique aspect of humans (and higher primates), and this is related to its position on the hand. The thumb axis has its foundation at the trapeziometacarpal joint and is normally pronated and flexed approximately 80° with respect to the other metacarpals of the hand.[1] This unique position allows for circumduction of the thumb, which then facilitates opposition of the thumb to the digits.

Thumb opposition is required for all useful prehension, and its preservation is needed as the basis for many hand surgical procedures. Thumb opposition results from the angular and rotatory motion produced via palmar abduction at the trapeziometacarpal joint as well as flexion and rotation of the trapeziometacarpal and the MCP joints.[2]

Functional opposition requires recruitment of multiple muscle groups. These muscles include the abductor pollicis brevis, opponens pollicis, and the superficial head of the flexor pollicis brevis. These muscles work simultaneously on the trapeziometacarpal joint and the MCP joint. The major force of opposition comes from the abductor pollicis brevis, with the opponens pollicis and flexor pollicis brevis providing secondary motors for opposition maneuvering. The extensor

Fig. 4. Chuck grip/directional grip.

Fig. 5. Hook grip.

pollicis longus and the adductor pollicis are antagonistic to thumb opposition and provide supination, extension, and adduction forces to thumb maneuvering.

The complex motions of the thumb are facilitated by the coordination of intrinsic thenar and extrinsic muscle groups. The thumb muscles allow for precision pinching and power gripping, and its stability is maintained actively by muscles and not by articular constraints. Opposition of the thumb involves the combined motions of flexion, pronation, and palmar adduction of the thumb metacarpal. The opposite of opposition (reposition) is performed by extension, supination, and adduction of the thumb metacarpal.

The abductor pollicis brevis is a subcutaneous muscle that lies radial to the flexor pollicis brevis. It originates primarily from the transverse carpal ligament, with some fibers coming from the scaphoid tubercle and the trapezium. The muscle then inserts onto the radial base of the thumb proximal phalanx, with some fibers also going to the radial side of the MCP joint. It should be noted that its fibers do blend with those of the flexor pollicis brevis. The recurrent branch of the median nerve innervates this muscle. The vascular supply of the abductor pollicis brevis is from the superficial radial ulnar branch of the radial artery. The abductor pollicis brevis can have variation in its anatomy, including additional heads and varying attachments. The main function of the abductor pollicis brevis is abduction and flexion of the thumb metacarpal. This abduction and flexion results in the thumb being pulled away from the palm at a right angle, initiating the act of opposition. The muscle also functions to extend the thumb IP joint through its extensor pollicis longus insertion and ulnarly deviates the MCP joint.[3–6]

The opponens muscle is a short and thick muscle that lies beneath the abductor pollicis brevis. It originates from the carpal metacarpal joint capsule as well as the tubercle of the trapezium and the transverse carpal ligament and inserts onto the volar radial aspect of the thumb metacarpal. Its vascular supply is via the superficial palmar branch of the radial artery. It has innervation from the recurrent branch of the median nerve in most cases but can also have dual innervation or just ulnar innervation.[5–9] The opponens flexes and pronates the thumb metacarpal. The opponens helps magnify the forces of opposition generated by the abductor pollicis brevis.[10]

The flexor pollicis brevis has a superficial (lateral) and a deep (medial) head. The superficial head has its origin from the tubercle of the trapezium and the transverse carpal ligament and inserts onto the radial base of the thumb proximal phalanx. The deep head rises from the trapezoid, capitate, and volar ligament of the distal carpal row and inserts onto the radial sesamoid and the base of the proximal phalanx. It receives its innervation from the recurrent motor branch of the median nerve and its vascular supply from the superficial palmar arch branch of the radial artery. In most cases, the superficial head muscle innervated by the recurrent branch of the median nerve and the deep head is most commonly innervated

Fig. 6. Power grasp.

Fig. 7. Span grasp.

by the deep motor branch of the ulnar nerve.[11] The Riche-Cannieu anastomosis is a nerve branch between the deep motor branch of the ulnar nerve and the recurrent motor branch of the median nerve and has been described as occurring in up to 77% of dissections.[12] The primary action of the flexor pollicis brevis is to help flex the MCP joint, as well as extend the distal phalanx, and pronate the thumb metacarpal.[5,6,10,13] It is noted that the thumb does not have a lumbrical, and this is thought (teleologically) to be because the thumb has no PIP joint, thus, no mechanical need for a lumbrical. It also has a highly mobile carpal metacarpal joint that allows for substantial 3-dimensional positioning.

The abductor of the thumb also has 2 heads, an oblique and a transverse. The oblique head has its origin on the capitate and the bases of the second and third metacarpals as well as the volar metacarpal ligaments and, in some cases, on the sheath of the flexor carpi radialis tendon. Most of its fibers unite to converge with the tendons of the flexor pollicis brevis (deep head), and the transverse head of the adductor inserts on the ulnar base of the thumb proximal phalanx and the dorsal extensor apparatus. This muscle tendon unit does have a sesamoid present. Another group of fibers can coalesce beneath the tendon of the flexor pollicis longus to join the deep head of the flexor pollicis brevis. The first dorsal interosseous muscle lays on the dorsum of the adductor, and the 2 muscles make up the bulk of the first web space. The deep motor branch of the ulnar nerve mainly innervates the adductor, and it gets its blood supply from the princeps pollicis artery. The adductors main action is adduction of the thumb metacarpal. It also helps with the extension of the thumb IP joint.

In considering reconstruction of the oppositional pinch, tendon transfers are an option. Cooney and colleagues[14] studied the muscle cross-sectional area and muscle forces to determine the ideal donor muscle for oppositional force recreation via transfer. The flexor digitorum superficialis (FDS) of the long finger and the extensor carpi ulnaris muscles were closely approximated to thenar muscle strength and potential excursion. However, abduction from the palm was greatest after a transfer of the FDS from the long and ring fingers and after extensor carpi ulnaris and extensor carpi radialis longus transfers. The motion and strength of the transfers were influenced by properly locating them within the hand.[15] Cooney and colleagues[14] emphasized the importance of directing the force of the transfer toward the pisiform. They noted that transfers distal to the pisiform, such as those using extensor digiti minimi or abductor digiti minimi, produced greater amounts of flexion than abduction. Transfers proximal to the pisiform, such as those using FDS and flexor carpi ulnaris, produce more abduction force and less metacarpal flexion.

The trapeziometacarpal joint is very complex because of its inherent instability and the fact that it is on the radial aspect of the wrist with no bony stabilizers. There are 5 major internal ligamentous stabilizers of the trapeziometacarpal joint:

1. The dorsal radial ligament
2. The posterior oblique ligament
3. The first intermetacarpal ligament
4. Ulnar collateral ligament
5. Anterior oblique ligament

The dorsal radial ligament prevents lateral subluxation. The posterior oblique ligament provides stability in flexion, opposition, and pronation. The first metacarpal is held tightly against the second metacarpal by the first intermetacarpal ligament, which is firm in abduction, opposition, and supination. The ulnar collateral ligament joins the intermetacarpal ligament in preventing lateral subluxation of the first metacarpal on the trapezium and helps control rotational forces. The fifth ligament, and the most important according to some investigators, is the volar anterior oblique ligament, which has both deep and superficial fibers. The ligament originates from the volar tubercle of the trapezium and it attaches on the volar aspect of the thumb metacarpal. This ligament is taut in extension, abduction, and pronation. Its biomechanical purpose is to control pronation forces and mitigate radial translation. The deep aspect of the anterior oblique ligament serves as a pivot point for the trapeziometacarpal joint and guides the metacarpal into pronation while thenar muscles work to produce abduction and flexion. These fibers help control ulnar translocation of the metacarpal during palmar abduction while the superficial anterior oblique ligament controls volar subluxation of the metacarpal.

THE FINGERS

The index finger is probably the next most important to the hand because of its ability to abduct and adduct, its ability to flex and extend, and its proximity to the thumb. Several studies have noted its importance in precision pinch and directional grip.[16–19] Murray and colleagues[20] have studied the loss of the index finger; they found that key pinch, power grip, and supination strength were diminished by approximately 20% after the loss of the index finger. In the normal hand, the width

of the grip extends from the hypothenar region all the way to the index finger. The radial aspect of the palm represents the external fulcrum of movement, and the ulnar aspect represents the internal fulcrum. In this study, those that did not have any dysesthetic pain thought that their overall hand function had improved with removal of the compromised index finger.

The long finger provides the most individual flexion force.[21,22] The central position means that it participates in both power grip and precision movements. The ring finger is noted to have less strength than either the long or the index finger and is uncommonly used for precision grip or pinch maneuvers. Tubiana and colleagues[16] thought that the loss of the ring finger resulted in the least amount of impairment to the hand.

The small finger has the least strength in flexion; however, its loss can result in decreased ability to hold objects in the palm. Part of the small finger's uniqueness is its carpal metacarpal joint, which can move approximately 25° in most people. Also, the hypothenar muscles add stabilization, which augments the flexion of the proximal phalanx of the small finger. The small finger's ability to abduct also helps enhance grasping while spanning an object. Tubiana thought the small finger was of a great impairment value, second only to the thumb.[16]

For finger range of motion, 15% of the intrinsic digital flexion occurs at the DIP joint but the DIP joint only contributes 3% to the overall flexion arc of the finger.[23] The PIP joint contributes 85% of the intrinsic digital flexion and adds 20% to the overall arc of the finger motion.

MCP joints, in many investigators' opinions, represent the most important joint for hand function because they contribute 77% of the total arc of finger flexion.[23–27] The DIP and PIP joints are ginglymoid-type joints that function more like hinges. The MCP joint is diarthrodial, which allows for flexion extension as well as abduction and adduction; they also allow for mild to moderate rotation of the digit.[26,28–30] Most prehension grips require that the digits be able to extend and abduct at the MCP joint. In order to be able to perform a precision pinch, the hand must have functional rotation and ulnar deviation at the MCP joint.[28,29] In order to accomplish a pinch, the radial intrinsic muscles and the collateral ligament to the index finger must hold against the stress applied by the thumb.

LUMBRICALS

Lumbricals are small muscles that are unique in that they arise on their antagonist. Their origin is the flexor digitorum profundus, and their main insertion point is the extensor expansion. The lumbricals contribute to MCP joint flexion, and they assist in IP joint extension. The lumbrical muscle name originates from the Latin word for worm. These 4 cylindrical-shaped muscles are located in the mid palm. Muscles have their origin on the flexor digitorum profundus tendons in the course along the palm in an almost parallel fashion to insert on the radial side of the digits. The most common pattern seen is that the first and second lumbricals originate from the radial side of the index and long finger deep flexors, and the ring and small finger lumbricals originate as bipennate muscle bellies on the adjacent surfaces of the flexor digitorum profundus tendons.[31,32] However, there have been numerous anatomic variants that have been described, and the hand surgeon should be familiar with some of these variants. In general, the muscles tend to follow a pattern of increasing variability as one moves from the radial aspect of the hand to the ulnar aspect.

Lumbricals can also have variable insertions, including the proximal phalanx; the volar plate of the MCP joint; and the extensor apparatus, including the lateral band and transverse oblique fibers of the extensor hood. Even though the lumbricals do insert onto the lateral band, other insertions onto other oblique fibers or transverse fibers have been found in more than 50% of specimens dissected. Furthermore, almost 50% have volar plate and/or bony attachments. Only one-quarter of the muscles went only to the radial lateral band as is classically described.[33]

The lumbrical muscles have very unique biomechanical properties. Their muscle mass and cross-sectional area are lowest in the upper extremity; however, their ratio between fiber and muscle length is largest in the upper extremity. These properties show that the lumbricals are really made for high excursions and that muscle contractile forces are constant over a wide range of fiber lengths.[34] Innervation of the first and second lumbricals is by the median nerve, whereas the third and fourth lumbricals are usually innervated by the ulnar nerve. Of note is that when using lumbricals as a proximally based pedicle flap, the radial-sided lumbricals are a safer option because of the comparatively fewer variations in the origin of the vascular supply and nerve supply.

The lumbrical muscle helps the extensor apparatus. Stimulation of the lumbrical produces IP joint extension followed by metacarpal joint flexion. As classically described, the lumbrical arises from the flexor digitorum profundus tendon; because of this, it is the only muscle that can relax the tendon of its antagonist. When considering

lumbrical function, it is best to remember its 2 attachments to the profundus tendon and to the lateral band. Flexion of the proximal phalanx is mainly caused by interossei. However, if the interossei are paralyzed, the lumbrical can initiate flexion at this joint. The lumbrical muscles are richly supplied with muscle spindles. Their passive elongation during contraction of the flexor digitorum profundus is thought to inhibit digit extensor groups and facilitate wrist extensors.[35–38] Because of this, lumbrical muscles have been called tension meters between the flexors and extensors.[39] Leijnse and Kalker[40] thought that the lumbricals provide proprioceptive feedback regarding the PIP and DIP joint motion and positions. Other investigators[41] have thought that their unique properties indicate they are important for fast and sudden changing movements and other fine movements of the hand.

A lumbrical plus finger results when the profundus tendon is lacerated and the normal tone of the muscle belly puts the cut end of the attachment of the lumbrical in a more proximal position. This proximal position causes increased tension on the radial lateral band, and the PIP joint may extend or hyperextend as the finger is actively flexed. This paradoxic extension is known as the lumbrical plus finger[42]; in cases of distal amputation, which severs the profundus tendon, a similar issue can arise. Furthermore, if a flexor tendon graft is placed too loosely, a lumbrical plus finger can result. As patients try to flex IP joints, the profundus pulls first on the lumbrical rather than on the loose tendon graft; this causes paradoxic extension of the PIP joint.

Numerous investigators[43–53] have suggested that carpal tunnel syndrome can be caused by anomalous lumbrical origins as well as hypertrophy of lumbrical muscles. The muscle can be potentially found within the canal resulting in subsequent compression of the median nerve. Carpal tunnel syndrome tends to spare the innervations of the lumbricals because their innervation is in a more dorsal position and better protected from direct compression.[54]

THE INTEROSSEI

The interosseous muscles of the hand are innervated by the ulnar nerve and organized into dorsal and palmar layers. These muscles have a small excursion but a great impact on finger balance, grip, and pinch ability. Their importance is unfortunately only appreciated after denervation and/or contracture, which result in impairment of the hand. The 3 palmar interossei adduct the fingers, and the 4 dorsal interossei abduct the fingers relative to the midline of the hand. The unique anatomy of the interossei muscles and their insertions and actions allow both sets of muscles to work in synergy. All of the interossei originate from metacarpal shafts. The palmar muscles originate from the ulnar side of the first and second metacarpals and the radial side of the fourth and fifth metacarpals, which positions them nicely for adduction of the fingers. The origin of the dorsal group is from the opposite surfaces of adjacent metacarpal shafts, which allows for finger abduction. The 3 palmar interosseous muscles insert distal to the MCP joint, into the extensor expansion of the finger. The dorsal interossei typically have both a bony insertion to the base of the proximal phalanx and the soft tissue insertion into the extensor aponeurosis. The interossei are key intrinsic balancers of the fingers, acting as flexors of the MCP joints and extensors for the IP joints. The dynamic actions of the interossei help stabilize the hand in the intrinsic plus position. The isometric actions of these muscles help stabilize the fingers throughout their dynamic movements.

It is important to remember that there are fascia boundaries around the interosseous muscles; because of this, compartment syndrome can occur within the interosseous compartments. Because there are both dorsal and palmar muscles, there are fascia layers that enclose both sets of interosseous muscles. These layers need to be released when performing fasciotomies of the hand.

The importance of the interossei on grip and pinch strength has been demonstrated by Kozin and colleagues.[55] They showed decreased grip strength of 38% and a 77% decrease in key pinch strength by measuring them after an ulnar nerve block in otherwise healthy individuals. This article by Kozin and colleagues[55] also showed that each median and ulnar innervated muscle had approximately 40% contribution to overall grip strength. There is a contribution from the interossei and lumbricals to IP extension and MCP flexion.

In high and low ulnar nerve palsies as well as crushing injuries, skin contracture and median nerve injury can all lead to failure and loss of function of the interosseous muscles. Ulnar nerve injury results in disruption of the finger balance with an intrinsic minus deformity: MCP joints hyperextended and the PIP and DIP joints flexed. This injury results because the interossei are mainly flexors of the MCP joints and extensors of the PIP joints. The extrinsic extensors are not strong PIP joint extensors. Ischemic injury as well as prolonged casting can lead to contracture of the interossei with resultant stiff hands and fingers, which may be in either the ulnar plus or ulnar minus position. The optimal positioning of splinting or casting

of the hand and wrist keeps the MCP joints flexed approximately 60° and leaves the PIP joints free to move through a flexion and extension arc, which allows for interosseous stretching.

THE HYPOTHENAR MUSCLES

The hypothenar muscles, as a group, originate partially from the transverse carpal ligament, partially from the volar carpal ligament, and partially from the adjacent carpal bones. There has been some concern, theoretically, that a transverse carpal ligament release could result in hypothenar muscle shortening with subsequent grip strength weakness. The abductor digiti minimi muscle originates from the pisiform and the flexor carpi ulnaris tendon as well as the pisohamate ligament. In most cases, it has 2 muscle slips with one inserting onto the ulnar aspect of the small finger proximal phalanx base and the other inserting into the extensor apparatus of the small finger. The main purpose of the abductor digiti minimi is small finger abduction. It also has a small contribution in finger MCP joint flexion and IP joint extension.[56] The flexor digiti minimi muscle has its origin on the hook of the hamate and the ulnar aspect of the flexor retinaculum. It fuses distally with the muscle of the abductor digiti minimi. The flexor digiti minimi muscle has a relatively large moment arm and is responsible for small finger MCP joint flexion. However, it does also have a component of small finger abduction. The biomechanical forces of the flexor digiti minimi and abductor digiti minimi rely on the pisiform. Since Because the pisiform is mobile, the relative position of this bone influences the action of the aforementioned muscles.

The opponens digiti minimi muscle has 2 layers with separate origins. The superficial layer has its origin on the hook of the hamate and inserts onto the distal ulnar aspect of the fifth metacarpal. The deep layer has its origin from the ulnar flexor compartment wall and inserts onto the proximal ulnar aspect of the small finger metacarpal shaft. The opponens digiti minimi muscle is deep to the 2 hypothenar muscles and is separate from them. When the metacarpal is flexed by the hypothenar muscles, the flexor carpi ulnaris contracts to stabilize the pisiform; this results in the ulnar arch becoming further flexed in a position that is otherwise known as *cupping*.

The 3 muscles of the hypothenar eminence also provide a substantial soft tissue envelope over the ulnar aspect of the hand that helps absorb impacts and other trauma that the hand may encounter. In many ways, it can be viewed that these muscles function as shock absorbers. Almost instinctively people will use the ulnar aspect of the hand with the hypothenar muscles to hammer or apply blunt force to an object, which can be seen in such activities as martial arts. The ulnar artery and its branches provide the vascular supply to these muscles. Specifically, the deep palmar branch of the ulnar artery seems to supply substantial amounts to these muscle groups. The ulnar nerve supplies the innervation specifically through a deep branch of the ulnar nerve, which travels through the canal and around the hook of the hamate where it provides motor innervation to the hypothenar muscles. It is noted that there can be significant variation in the motor nerve distribution in the hypothenar muscles making surgery challenging in this area.

TENDONS

The extensor tendon system has less movement or excursion than the flexor tendon system does.[57] The extensor apparatus also has decreased ability to compensate for shortening because of the connection between the intrinsic and the extrinsic mechanisms of this tendon system. Excursion of the extensor tendon at the level of the PIP joint is only 2 to 5 mm. The profundus flexor tendon provides a terminal pinch. Loss of the flexor profundus tendon may prevent full digital palmar grip.[58,59] Even though the quadralgia effect classically only applies to the long, ring, and small fingers because of the common muscle belly, the quadralgia effect can also involve the index finger because of the copious synovium present at the level of the carpal tunnel. This tissue has been called the *fibromembranous retinaculum*, which can link the index profundus tendon to the long, ring, and small finger tendons. The flexor superficialis tendon helps provide balance to the finger flexion arc. Loss can result in hyperextension at the PIP joint. Even with loss of both flexor tendons, approximately 45° of flexion can be possible at the MCP joints via the intrinsics. The lumbrical plus deformity is caused by the contracted profundus muscle belly placing stretch on the shortened lumbrical while the finger is flexing, this results in a paradoxic extension of the PIP joint. This interesting biomechanical effect can be solved by dividing the lumbrical or by suturing the profundus tendon to the flexor tendon sheath in a relaxed position.[58,60] Of course, during flexor tendon surgery, it is imperative to try and preserve the A2 and A4 pulleys. If these cannot be preserved or have been injured, then repair or reconstruction is needed. The A2 and A4 pulleys are located over the central portions of the proximal and middle phalanges. This

location allows them to help prevent bowstringing that would occur with joint flexion if the pulleys were not present. The A1, A3, and A5 pulleys have a variable relationship to the joint axis and are only helpful in restraining some bowstringing. The cruciate pulleys vary in their location and contribute little biomechanical resistance to bowstringing.[61,62]

THE HAND AND WRIST

In order to perform power grip, a stable wrist is needed. Biomechanically, a stable wrist prevents the dissipation of finger flexion and extensor forces as the tendons move over the carpus. The human hand is one of the most complex biomechanical systems. It is a system of bony segments that are arranged in a series of longitudinal and transverse arches.[63] Essentially, there are 2 transverse arches: the proximal transverse arch formed by the carpal bones and the distal transverse arch formed by the metacarpal heads of the fingers. The longitudinal arches are made up of the bones of the 5 digital rays. The proximal aspect of the longitudinal arches and the proximal transverse arches converge at the carpal bones. Therefore, the carpal bones contribute stabilizing components to the longitudinal arches as well as the related structures central to hand function. Biomechanically, the arch resists greater force than other structures, and that is why arches are frequently found throughout ancient and modern architecture and engineering.

Carpal bones help stabilize the motion of the hand and make up part of the joint motion arc. It is well known that the Fibonacci ratio of 1:1.618034 is found in the lengths of the metacarpal, proximal phalanx, middle phalanx, and distal phalanx bones. Other investigators have noted[64,65] that the functional lengths of these bones using the center of rotation about the joints fits the ratio better than absolute bone lengths. This ratio is found in the nautilus shell, sunflowers, eggshells, spiral galaxies of outer space, and the Parthenon in Greece. This geometric design, in this ratio, and the bony structures of the hand make an equiangular spiral of joint motion arcs.

SUMMARY

The biomechanics of the hand are truly amazing. In many ways, the complexity of the hand defies our ability to fully comprehend the marvel of evolutionary engineering that has resulted in its design. Injuries to the bones, tendons, and/or ligaments of the hand can result in permanent loss of function and significant impairment. The goal of the hand surgeon is to mitigate this functional loss by restoring the anatomy when possible and fooling Mother Nature when needed.

REFERENCES

1. Napier JR. The form and function of the carpometacarpal joint of the thumb. J Anat 1955;89:362.
2. Cooney WP, Lucca MJ, Chao EY, et al. The kinesiology of the thumb trapeziometacarpal joint. J Bone Joint Surg Am 1981;63:1371–81.
3. Botte MJ. Muscle anatomy. In: Doyle JR, Botte MJ, editors. Surgical anatomy of the hand and upper extremity. Philadelphia: Lippincott Williams & Wilkins; 2003. p. 149–55.
4. Standring S, editor. Gray's anatomy. 40th edition. Elsevier limited; 2008.
5. Leversedge FJ. Anatomy and pathomechanics of the thumb. Hand Clin 2008;28(4):681–4.
6. Leversedge F, Goldfarb C, Boyer M. Chapter 1, hand; Chapter 5, neuroanatomy. In: Leversedge F, Goldfarb C, Boyer M, editors. A pocket manual of hand and upper extremity primus manus. Philadelphia: Wolters Kluwer Lippincott Williams &Wilkins; 2010. p. 24–9.
7. Rowntree T. Anomalous innervation of the hand muscles. J Bone Joint Surg Br 1949;31(4):505–10.
8. Day MH, Napier JR. The two heads of flexor pollicis brevis. J Anat 1961;95:123–30.
9. Ajmani ML. Variations in the motor nerve supply of the thenar and hypothenar muscles of the hand. J Anat 1996;189(Pt 1):145–50.
10. Kaplan EB, Smith RJ. Kinesiology of the hand and wrist and muscular variations of the hand and forearm. In: Spinner M, editor. Kaplan's functional and surgical anatomy of the hand. 3rd edition. Philadelphia: J.B. Lippincott; 1984. p. 283–349.
11. Beasley RW. Surgical anatomy of the hand. In: Beasley RW, editor. Beasley's surgery of the hand. New York: Thieme; 2003. p. 18.
12. Harness D, Sekeles E. The double anastomotic innervation of thenar muscles. J Anat 1971;109(Pt 3):461–6.
13. Morris's human anatomy, part II: a complete systematic treatise by English and American authors. 4th edition. Philadelphia: Blakiston's Son & Co; 1907. p. 395–8 *Book digitized by Google from the library of Harvard University and uploaded to the Internet Archive.
14. Cooney WP, Linscheid RL, An KN. Opposition of the thumb: an anatomic and biomechanical study of tendon transfers. J Hand Surg Am 1984; 9:777–86.
15. Bunnell S. Opposition of the thumb. J Bone Joint Surg 1938;20:269–84.
16. Tubiana R, Thomine J, Mackin E. Movements of the hand and wrist. In: Tubiana R, Thomine J,

Mackin E, editors. Examination of the hand and wrist. St Louis (MO): Mosby; 1996. p. 40–125.
17. Duparc J, Alnot JY, May P. Single digit amputations. In: Campbell DA, Gosset J, editors. Mutilating injuries of the hand. Edinburgh (United Kingdom): Churchill Livingstone; 1979. p. 37–44.
18. Buck-Gramcko D, Hoffmann R, Neumann R, editors. Hand trauma: a practical guide. New York: Theime; 1986. p. 60–73.
19. Campbell DA, Gosset J, editors. Mutilating injuries of the hand. Edinburgh (United Kingdom): Churchill Livingstone; 1979. p. 37–44.
20. Murray JF, Carman W, MacKenzie JK. Trans- metacarpal amputation of the index finger: actual assessment of hand strength and complications. J Hand Surg 1977;2:471–81.
21. Ejeskar A, Ortengren R. Isolated finger flexion force: a methodological study. Hand 1981;13:223–30.
22. Hazelton FT, Smidt GL, Flatt AE, et al. The influence of wrist position on the force produced by the finger flexors. J Biomech 1975;8:301–6.
23. Littler JW, Herndon JH, Thompson JS. Examination of the hand. In: Converse JM, Littler JW, editors. Reconstructive plastic surgery, vol. 6. Philadelphia: WB Saunders; 1977. p. 2973.
24. Littler JW, Thompson JS. Surgical and functional anatomy. In: Bowers WH, editor. The interphalangeal joints. New York: Churchill Livingstone; 1987. p. 142.
25. Foucher G, Hoang P, Citron N, et al. Joint reconstruction following trauma: comparison of microsurgical transfer and conventional methods: a report of 61 cases. J Hand Surg Br 1986;11:388–93.
26. Ellis PR, Tsai T. Management of the traumatized joint of the finger. Clin Plast Surg 1989;16:457–73.
27. Swanson AB. Flexible implant arthroplasty for arthritic finger joints: rationale, technique, and results of treatment. J Bone Joint Surg Am 1972;54: 435–55.
28. Beckenbaugh RD, Dobyns JH, Linscheid RL, et al. Review and analysis of silicone-rubber metacarpophalangeal implants. J Bone Joint Surg Am 1976; 58:483–7.
29. Flatt AE. Care of the rheumatoid hand. 4th edition. St Louis (MO): Mosby; 1983.
30. Krishnan J, Chipchase L. Passive and axial rotation of the metacarpophalangeal joint. J Hand Surg Br 1997;22:270–3.
31. Shin YA, Amadio PC. Stiff finger joints. In: Green DP, Hotchkiss RN, Pederson WC, et al, editors. Green's operative hand surgery. 5th edition. Philadelphia: Elsevier; 2005. p. 422–3.
32. Goldberg S. The origin of the lumbrical muscles in the hand of the South African native. Hand 1970;2: 168–71.
33. Eladoumikdachi F, Valkov PL, Thomas J, et al. Anatomy of the intrinsic hand muscles revisited: part II. Lumbricals. Plast Reconstr Surg 2002; 110(5):1225–31.
34. Jacobson MD, Raab R, Fazeli BM, et al. Architectural design of the human intrinsic hand muscles. J Hand Surg Am 1992;17:804–9.
35. Ranney D, Wells R. Lumbrical muscle function as revealed by a new and physiological approach. Anat Rec 1988;222(1):110–4.
36. Buford WL Jr, Koh S, Andersen CR, et al. Analysis of intrinsic-extrinsic muscle function through interactive 3-dimensional kinematic simulation and cadaver studies. J Hand Surg Am 2005;30(6): 1267–75.
37. Backhouse KM, Catton WT. An experimental study of the function of the lumbrical muscles in the human hand. J Anat 1954;88(5):133–41.
38. Devanadan MS, Ghosh S, John KL. A quantitative study of the muscle spindles and tendon organs in some intrinsic muscles of the hand. Anat Rec 1983;207:263–6.
39. Rabischong P. Basic problems in the restoration of prehension. Ann Chir 1971;25(19):927–33.
40. Leijnse JN, Kalker JJ. A two-dimensional kinematic model of the lumbrical in the human finger. J Biomech 1995;28(3):237–49.
41. Leijnse JN. Why the lumbrical muscle should not be bigger – a force model of the lumbrical in the unloaded human finger. J Biomech 1997;30(11–12): 1107–14.
42. Smith RJ. Intrinsic muscles of the fingers: function, dysfunction and surgical reconstruction. In: AAOS instructional course lectures, vol. 24. St Louis (MO): Mosby; 1975. p. 200–20.
43. Butler B Jr, Bigley EC Jr. Aberrant index lumbrical tendinous origin associated with carpal tunnel syndrome: a case report. J Bone Joint Surg 1971;53: 160–2.
44. Schultz RJ, Endler PM, Huddleston HD. Anomalous median nerve and an anomalous muscle belly of the first lumbrical associated with carpal tunnel syndrome. J Bone Joint Surg 1973;55:1744–6.
45. Still JM Jr, Kleinert HE. Anomalous muscles and nerve entrapment in the wrist and hand. Plast Reconsrt Surg 1973;52:394–400.
46. Jabaley ME. Personal observations on the role of the lumbrical muscles in carpal tunnel syndrome. J Hand Surg 1978;3:82–4.
47. Erikson J. A case of carpal tunnel syndrome on the basis of abnormally long lumbrical muscle. Acta Orthop Scand 1973;44:275–7.
48. Robinson D, Aghasi M, Halperin N. The treatment of carpal tunnel syndrome caused by hypertrophied lumbrical muscles. Scand J Plast Reconstr Surg 1989;23:149–51.
49. Gainer JV, Nugent GR. Carpal tunnel syndrome: report of 430 operations. South Med J 1977;70: 325–8.

50. Rothfleisch S, Sherman D. Carpal tunnel syndrome, biomechanical aspects of occupational occurrence and implications regarding surgical management. Orthop Rev 1978;7:107–9.
51. Smith EM, Sonstegard DA, Anderson WH Jr. Carpal tunnel syndrome: contribution of flexor tendons. Arch Phys Med Rehabil 1977;58:379–85.
52. Gelberman RH, Hergenroeder PT, Hargens AA, et al. The carpal tunnel syndrome: a study of canal pressures. J Bone Joint Surg 1981;63:380–3.
53. Skie M, Zeiss J, Ebraheim NA, et al. Carpal tunnel changes and median nerve compression during wrist flexion and extension seen by magnetic resonance imaging. J Hand Surg 1990;15:934–9.
54. Yates SK, Yaworski R, Brown WF. Relative preservation of lumbrical versus thenar motor fibres in neurogenic disorders. J Neurol Neurosurg Psychiatr 1981;44(9):768–74.
55. Kozin SH, Porter S, Clark P, et al. The contribution of the intrinsic muscles to grip and pinch strength. J Hand Surg Am 1999;24(1):64–72.
56. Brand PW, Hollister AM. Clinical mechanics of the hand. 3rd edition. St Louis (MO): Mosby Inc; 1999. p. 173–5.
57. Verdan CE. Primary and secondary repair of flexor and extensor tendon injuries. In: Flynn JE, editor. Hand surgery. Baltimore (MD): Williams & Wilkins; 1966. p. 220–75.
58. Smith P. Lister's the hand. London: Churchill Livingstone; 2002.
59. Verdan CE. Syndrome of the quadriga. Surg Clin North Am 1960;40:425–6.
60. Louis DS, Jebson PJL, Graham TJ. Amputations. In: Green DP, Hotchkiss RN, Pederson WC, editors. Green's operative hand surgery. 4th edition. New York: Churchill Livingstone; 1999. p. 48–75.
61. Hume EL. Panel discussion: flexor tendon reconstruction. In: Hunter JM, Schneider LH, Mackin EJ, editors. Tendon surgery in the hand. St Louis (MO): Mosby; 1987. p. 658–62.
62. Lin A, Amadio PC, An K, et al. Functional anatomy of the human digital flexor pulley system. J Hand Surg Am 1989;14:949–56.
63. Flatt AE. Biomechanics of the hand and wrist. In: Evarts CM, editor. Surgery of the musculoskeletal system. 2nd edition. New York: Churchill Livingstone; 1983. p. 311–29.
64. Hamilton R, Dunsmuir RA. Radiographic assessment of the relative lengths of the bones of the fingers of the human hand. J Hand Surg 2002;27:546–8.
65. Park AE, Fernandez JJ, Schmedders K, et al. The Fibonacci sequence: relationship to the human hand. J Hand Surg 2003;28:157–60.